MCMP117 · Fundamentals of Operating Systems

Midterm Exam --- Sample Paper

Istanbul Okan University · Vocational School · Electronical Communication Technology Programme

Instructor: Asst. Prof. Dr. Hasan Oğuz Date: [to be announced] Room: [to be announced] Duration: [90] minutes

Name, surname: _____ **Student ID:** _____

Signature: _____ **Seat / room:** _____

Part	A	B	C1	C2	C3	C4	C5	**Total**
Points	20	20	16	12	12	10	10	**100**
Score								

Rules. This paper covers Weeks 1–7. [Closed book; no notes, no phones — to be confirmed.] [A basic, non-programmable calculator is allowed — to be confirmed.] Write your answers in the boxes and show every step of a calculation (formula → numbers → result with unit). **Course conventions:** context-switch time 0; ties broken by arrival time, then by name; a smaller priority number means a higher priority; in Round Robin a process arriving at the moment a quantum expires enters the queue before the preempted process; TLB: EAT $= t_{\text{TLB}} + (2-\alpha)\,t_m$; holes are listed in address order and a request is carved from the beginning of the chosen hole; page frames start empty.

Part A --- Multiple choice (10 × 2 = 20 points)

Circle *one* answer.

A1. The first successful time-sharing system, demonstrated at MIT in 1961, was (a) Multics (b) CTSS (c) OS/360 (d) MS-DOS

A2. In which OS structure do all services run as one program in kernel mode, so that it is fast but a buggy driver can crash the whole system? (a) microkernel (b) layered (c) monolithic (d) none of these

A3. In the *child* process, `fork()` returns (a) the child's PID (b) 0 (c) the parent's PID (d) -1

A4. Which transition does *not* exist in the five-state process model? (a) ready → running (b) running → waiting (c) waiting → running (d) running → ready

A5. Which quantity does the CPU-scheduling algorithm influence most directly? (a) the CPU burst length (b) the I/O time (c) the waiting time (d) the size of the program

A6. Linux and Windows map user threads to kernel threads with the model (a) many-to-one (b) one-to-one (c) many-to-many (d) one-to-many

A7. If the Round Robin quantum is larger than every CPU burst, Round Robin behaves like (a) SJF (b) FCFS (c) priority (d) SRTF

A8. To use execution-time address binding, the system needs (a) a linker (b) a relocating loader (c) an MMU (d) compaction

A9. The TLB is (a) a small, fast cache of page-table entries (b) the page table itself (c) a list of free frames (d) a disk area for swapped pages

A10. A system is thrashing. The correct reaction of the OS is to (a) admit more processes (b) increase the time quantum (c) replace LRU by OPT (d) reduce the number of processes in memory

Part B --- Short answer (4 × 5 = 20 points)

B1. Describe the two classic views of an operating system (*extended machine* and *resource manager*). Give one example of time multiplexing and one of space multiplexing. **[5 pt]**

B2. Compare user-level and kernel-level threads: give one advantage and one disadvantage of each. What happens in the many-to-one model when one thread makes a blocking system call? **[5 pt]**

B3. Two threads use semaphores S and Q, both initialised to 1: the threads execute

```
P0: wait(S); wait(Q); ... signal(S); signal(Q);
P1: wait(Q); wait(S); ... signal(Q); signal(S);
```

Give an order of execution that leads to a deadlock, explain why, and state a simple rule that prevents it. **[5 pt]**

B4. Explain the difference between a hard link and a symbolic link. A user runs `ln a.txt h.txt`, `ln -s a.txt s.txt` and then `rm a.txt`. Can the data still be read through `h.txt`? Through `s.txt`? Why? **[5 pt]**

Part C --- Problems (60 points)

C1. **[16 pt]** Four processes arrive at a single CPU (times in ms):

Process	Arrival	Burst	Priority
A	0	6	3
B	2	5	2
C	3	2	1
D	5	4	4

For (a) non-preemptive SJF, (b) preemptive priority scheduling and (c) Round Robin with $q = 3$ ms, draw the Gantt chart and fill in the table. (d) Which of the three gives the smallest average waiting time?

(a)



Process	Finish	Turnaround	Waiting	Response
A				
B				
C				
D				
Average	—			

(b)



Process	Finish	Turnaround	Waiting	Response
A				
B				
C				
D				
Average	—			

(c)



Process	Finish	Turnaround	Waiting	Response
A				
B				
C				
D				
Average	—			

(d) _____

C2. **[12 pt]** (a) A process has base register `0x6000` and limit register `0x0A00`. Translate the logical addresses `0x09FF`, `0x0A00` and `0x0150`. Give the legal range of physical addresses. **[4 pt]**

(b) A segment table contains: segment 0: base 1400, limit 600; segment 1: base 6300, limit 400; segment 2: base 4300, limit 1100; segment 3: base 3200, limit 1000. Translate $\langle 2, 1099\rangle$, $\langle 1, 400\rangle$ and $\langle 3, 512\rangle$. **[4 pt]**

(c) A paged system has $t_{\text{TLB}} = 20$ ns and $t_m = 100$ ns. Compute the effective access time for a hit ratio of 0.8 and without a TLB. Which hit ratio gives an effective access time of 128 ns? **[4 pt]**

C3. **[12 pt]** For the reference string $1, 0, 4, 2, 1, 0, 1, 2, 3, 1, 2, 3$ and 3 initially empty frames, fill in the frame tables and count the page faults for FIFO, LRU and OPT.

FIFO **faults:** _____

ref	1	0	4	2	1	0	1	2	3	1	2	3
fr. 0												
fr. 1												
fr. 2												
fault												

LRU **faults:** _____

ref	1	0	4	2	1	0	1	2	3	1	2	3
fr. 0												
fr. 1												
fr. 2												
fault												

OPT **faults:** _____

ref	1	0	4	2	1	0	1	2	3	1	2	3
fr. 0												
fr. 1												
fr. 2												
fault												

C4. **[10 pt]** Memory has five holes (size@start, in KB, in address order): $650@200$, $450@1100$, $200@1800$, $550@2200$, $250@3000$. Processes A $= 270$, B $= 290$, C $= 330$ and D $= 510$ KB arrive in this order. For first-fit, best-fit and worst-fit, give the hole used by each process and the size of the hole that is left. Which requests cannot be satisfied, and why?

Request	First-fit	Best-fit	Worst-fit
A $= 270$			
B $= 290$			
C $= 330$			
D $= 510$			

C5. **[10 pt]** (a) A file consists of 300-byte records numbered from 0; the file system uses 1024-byte blocks. In which relative block, and at which offset inside it, does record 45 start? Does record 58 fit into a single block? **[4 pt]**

(b) How many times is each letter printed by the following code (output to a terminal), and how many processes exist at the end? **[3 pt]**

```
fork(); printf("X\n"); fork(); fork(); printf("Y\n");
```

(c) A context switch takes $s = 20\ \mu$s and the time quantum is $q = 5$ ms. What percentage of the CPU time is lost to switching if every process uses its full quantum? **[3 pt]**

— End of the sample paper. Good luck! —